

Discovery 1
copper wire
↑
copper plate

step ~2,000

Discovery 2
circuit
↑
wire + iron

step ~8,000

New goal
blueprint requires
50 circuits

step ~15,000

Required
recall full de-
pendency chain;
reason about bottlenecks;
adapt the layout

memory across >10,000 steps

Transformers: attention fades beyond context • SSMs: compressed into fixed-size state • MANNs: no causal model of what to store